Part 1. Board problems

	Relation and given rate	Answer
Example 1	$y = x^2$, $x = 3$, $\frac{dx}{dt} = 2$	$\frac{dy}{dt} = 12$
Example 2	$x^2 + y^2 = 25$, $(x, y) = (3, 4)$, $\frac{dx}{dt} = 2$	$\frac{dy}{dt} = -\frac{3}{2}$
Practice 1	$xy = 12$, $(x, y) = (3, 4)$, $\frac{dy}{dt} = -2$	$\frac{dx}{dt} = \frac{3}{2}$

Part 2. Word problems

Example 3

$V = \frac{4}{3}\pi r^3$, so $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 2$, $\frac{dr}{dt} = \frac{12\pi}{16\pi} = \boxed{\frac{3}{4}\text{ cm/s}}$.

Practice 2

$V = s^3$, so $\frac{dV}{dt} = 3s^2 \frac{ds}{dt}$. At $s = 2$, $\frac{ds}{dt} = \frac{-12}{3(2)^2} = \boxed{-1\text{ m/s}}$.

Example 4

$A = LW$, so $\frac{dA}{dt} = W\frac{dL}{dt} + L\frac{dW}{dt} = 5(3) + 8(-1) = \boxed{7\text{ cm}^2/\text{s}}$.

Practice 3

$A = \frac{1}{2}bh$, so $\frac{dA}{dt} = \frac{1}{2}(h\frac{db}{dt} + b\frac{dh}{dt}) = \frac{1}{2}(6(-2) + 8(3)) = \boxed{6\text{ cm}^2/\text{min}}$.

Part 3. More challenging models

Example 5

Let x be the bottom's distance from the wall and y the top's height. Then $x^2+y^2=100$ and $x\frac{dx}{dt}+y\frac{dy}{dt}=0$. At $x=6$, $y=8$.

(a) $\frac{dy}{dt}=-\frac{6}{8}(2)=-\frac{3}{2}$ ft/s: the top moves $\boxed{\frac{3}{2}\text{ ft/s downward}}$.

(b) $\frac{dx}{dt}=-\frac{8}{6}(-3)=\boxed{4\text{ ft/s}}$, away from the wall.

Practice 4

Let x be the horizontal distance and L the rope length from bow to pulley. Then $L^2=x^2+9$, so $L\frac{dL}{dt}=x\frac{dx}{dt}$. At $x=4$, $L=5$ and $\frac{dL}{dt}=-1$, giving $\frac{dx}{dt}=-\frac{5}{4}$ m/s. The boat approaches at $\boxed{\frac{5}{4}\text{ m/s}}$.

Example 6

(a) Similar triangles give $\frac{r}{h}=\frac{2}{6}=\frac{1}{3}$. Thus $V=\frac{1}{3}\pi r^2h=\boxed{\frac{\pi}{27}h^3}$.

(b) $\frac{dV}{dt}=\frac{\pi h^2}{9}\frac{dh}{dt}$. At $h=3$, $\pi=\pi\frac{dh}{dt}$, so $\frac{dh}{dt}=\boxed{1\text{ m/min}}$.

(c) Draining gives $\frac{dV}{dt}=-\pi$, hence $\frac{dh}{dt}=\boxed{-1\text{ m/min}}$.

Practice 5

Let x be the person's distance from the light and s the shadow length. Similar triangles give $\frac{6}{x+s}=\frac{2}{s}$, so $s=\frac{x}{2}$.

(a) $\frac{ds}{dt}=\frac{1}{2}\frac{dx}{dt}=\boxed{1\text{ m/s}}$. (b) $\frac{d(x+s)}{dt}=2+1=\boxed{3\text{ m/s}}$.